

DEEP LEARNING — DL26

Knowledge Distillation for Mobile Action Recognition

Compressing a 3D ResNet-50 into an ultra-lightweight MobileNet3D on UCF-101
via logit-based, attention transfer, and cross-frame distillation.

STUDENT

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TRACK

Track 6 — KD for Mobile Action Recognition

INSTITUTION

Università degli Studi di Catania — DMI

ADVISOR

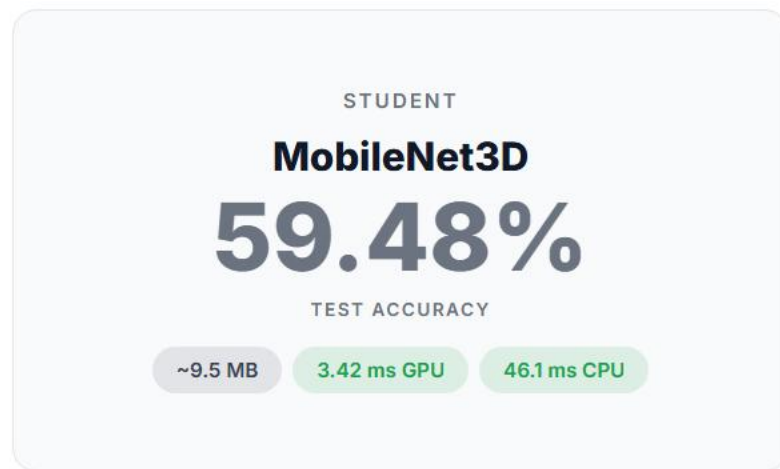
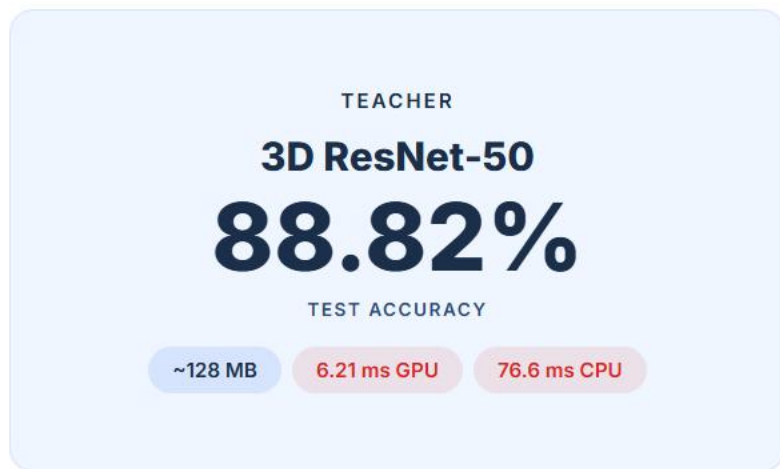
Prof. Antonino Furnari

REPOSITORY

github.com/danii909/KD_Project

The Capacity Gap

Goal: Compress a 3D ResNet-50 teacher (pretrained on Kinetics-400) into a MobileNet3D student for on-device video action recognition on UCF-101 (101 classes).



UCF-101 & Group-Aware Split

101

ACTION CLASSES

13,320

VIDEO CLIPS

24f

@ 112×112 PX

Original Split Structure

UCF-101 is officially divided only into Train and Test partitions, **lacking an actual validation set** required for proper model selection.

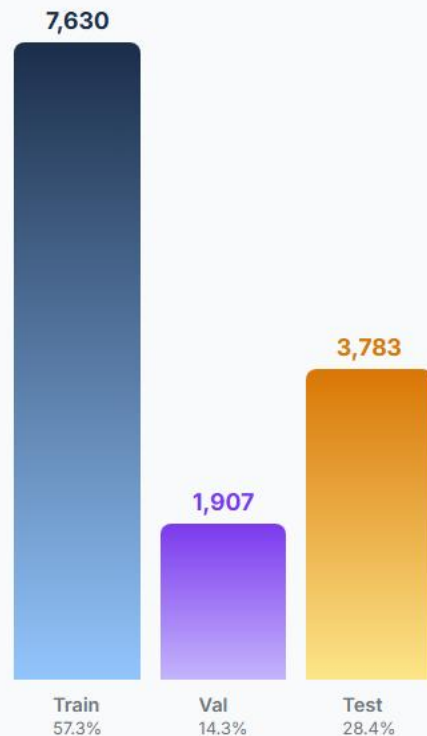
⚠ Data Leakage Risk

Correlated clips from the same source video share actor & background. Random splits cause **spatial leakage** and inflated accuracy.

✅ Group-Aware Split

Isolation by group ID (gXX). No source video shared across Train / Val / Test.

NEW SPLIT DISTRIBUTION



Augmentation Trade-off & Data Leakage

All tests: T=10, a=0.7, 24f · Why light augmentation was chosen

1. Minimal Augmentation

WITH LEAK

65.7%

Top-5: 87.8%

NO LEAK

59.3%

Top-5: 83.6%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Disabled (0.0)
- **Random Erasing:** Disabled
- **Temporal stride:** Fixed (max_stride=1)

2. Light Augmentation ✓

WITH LEAK

69.5%

Top-5: 90.3%

NO LEAK

64.5%

Top-5: 87.4%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Light (0.1)
- **Random Erasing:** 5% probability
- **Temporal stride:** Fixed (max_stride=1)

3. Strong Augmentation

WITH LEAK

55.83%

Top-5: 83.27%

NO LEAK

50.4%

Top-5: 78.5%

CONFIGURATION

- **Spatial crop:** Resized Crop (0.6—1.0)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Strong (0.2)
- **Random Erasing:** 15% probability
- **Temporal stride:** Variable (max_stride=2)

Temperature Sweep & Main Result

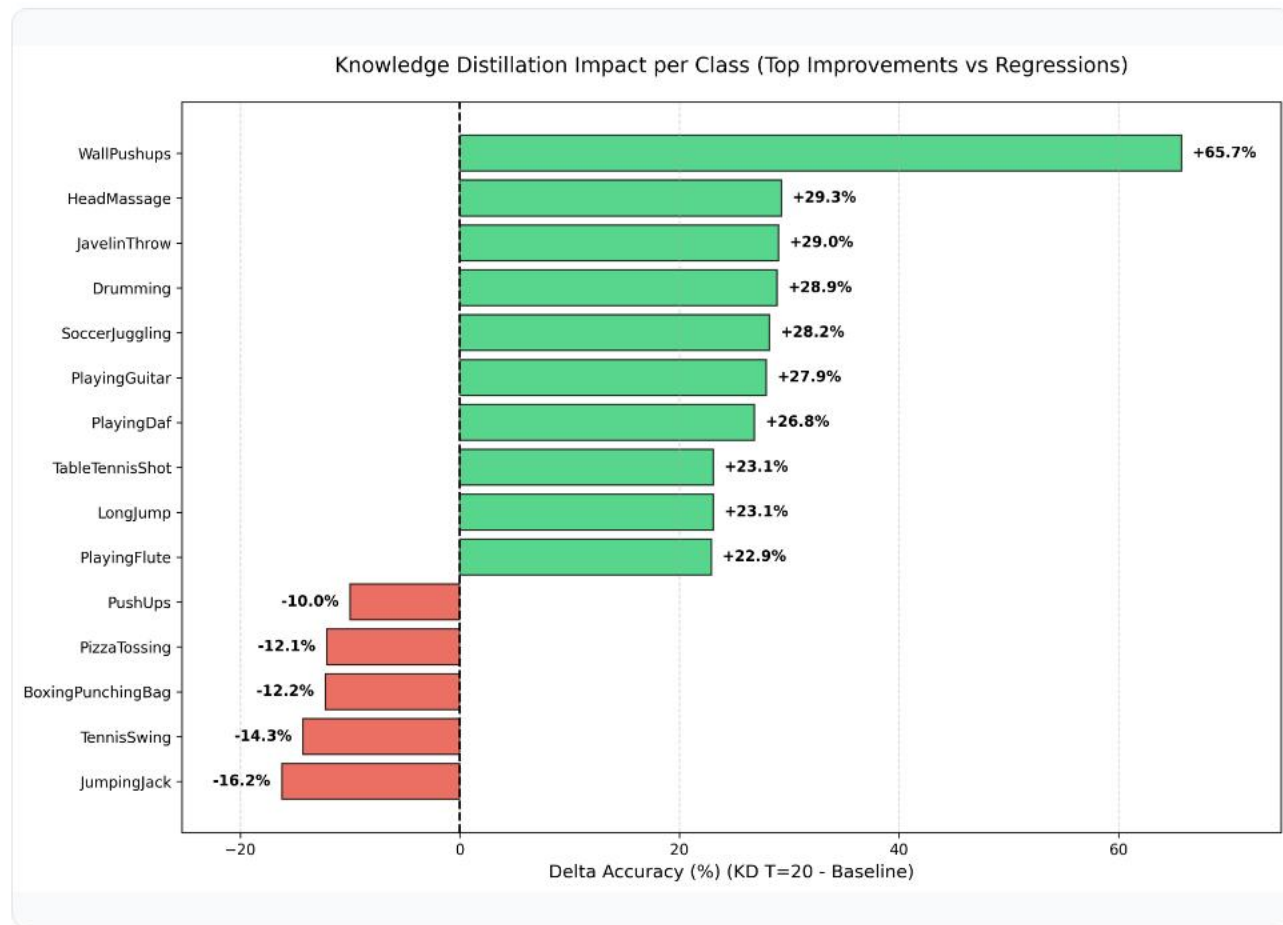
Systematic analysis of Student behavior as T varies · a = 0.7 fixed

TEMPERATURE	BEST VAL ACC	EPOCH	TRAIN ACC	TEST TOP-1	TEST TOP-5	Δ VS BASELINE
Minimal Augmentation (no KD)	59.18%	—	99.81%	59.48%	82.77%	ref.
T = 1	62.75%	60	98.44%	62.91%	85.59%	+3.43%
T = 5	63.12%	57	95.87%	62.07%	87.02%	+2.59%
T = 8	64.57%	56	97.59%	64.31%	87.68%	+4.83%
T = 10	64.85%	60	98.28%	64.47%	87.44%	+4.99%
T = 20	65.18%	59	99.05%	65.85%	87.81%	+6.37%

TEST ACCURACY VS. TEMPERATURE TREND



Per-Class Accuracy: KD T=20 vs Baseline



Improvements

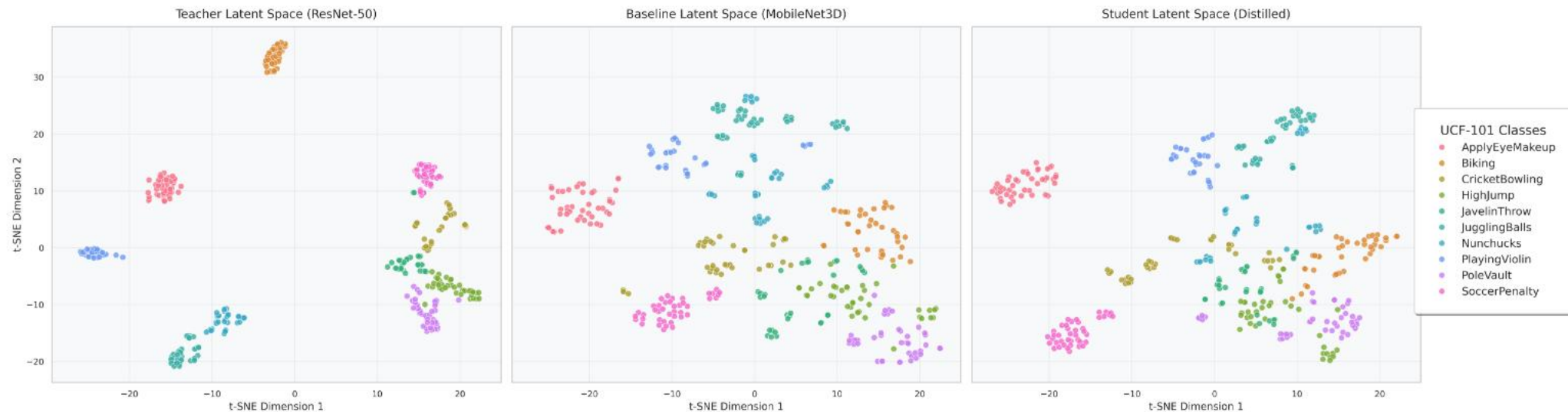
Spatially complex classes (e.g. **WallPushups**, **HeadMassage**) gain most from the Teacher's visual filters.

Regressions

Fast, cyclic actions (**JumpingJack**, **TennisSwing**) suffer from temporal over-smoothing at high T.

Latent Space Analysis via t-SNE

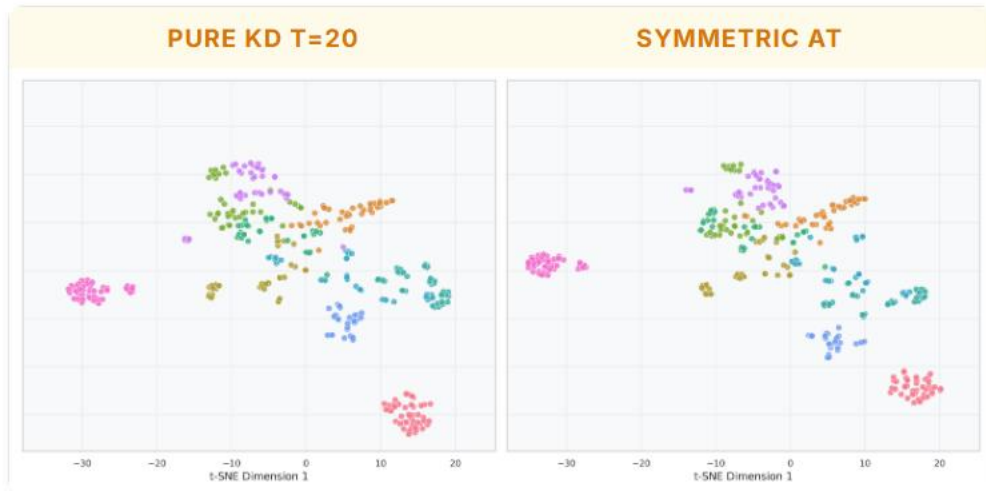
2D projections of pre-classifier embeddings — 10 selected classes



The distilled model (T=20) shows significantly tighter and more distinct class clusters compared to the baseline student, successfully inheriting the well-separated semantic boundaries of the teacher model.

Attention Transfer (AT)

Capacity gap impact on intermediate feature alignment



PURE KD T=20
65.85%

CAPACITY LOSS
-1.27%

SYMMETRIC AT
64.58%

Forced spatial alignment acts as noise due to Student's depthwise convolution capacity limit.



PURE KD T=20
65.85%

CAPACITY LOSS
-0.82%

TEMPORAL AT
65.03%

Removing spatial constraints ($\beta_s=0$) attenuates regression, as temporal dynamics are more transferable.

Born-Again Networks & Knowledge Collapse

Iterative self-distillation MobileNet3D → MobileNet3D

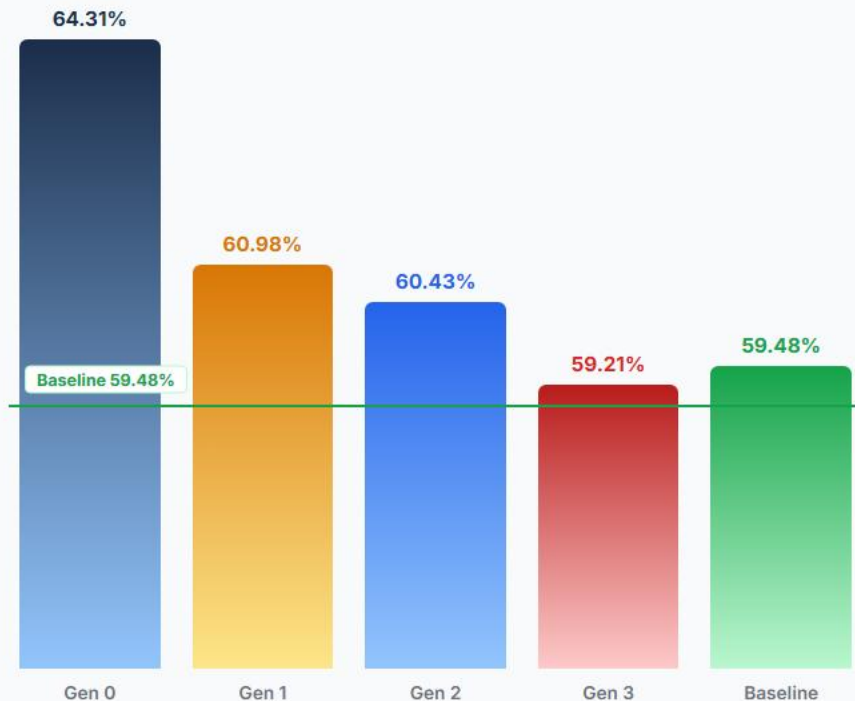
Iterative Self-Distillation

GEN	TEACHER	T	A	ACCURACY
Gen 0	3D ResNet-50	8.0	0.7	64.31%
Gen 1	MobileNet3D Gen 0	2.5	0.7	60.98%
Gen 2	MobileNet3D Gen 1	2.5	0.7	60.43%
Gen 3	MobileNet3D Gen 2	2.5	0.7	59.21%
Baseline	No Teacher (Scratch)	—	—	59.48%

Considerations

A compressed student (64.31%) lacks the visual representational capacity to act as an effective teacher. Distilling at $T=2.5$ fails to prevent error propagation, causing progressive degradation back to baseline (59.48%).

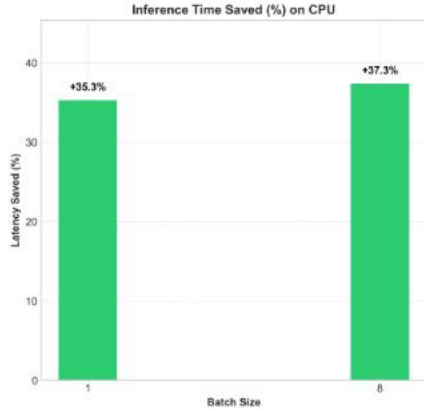
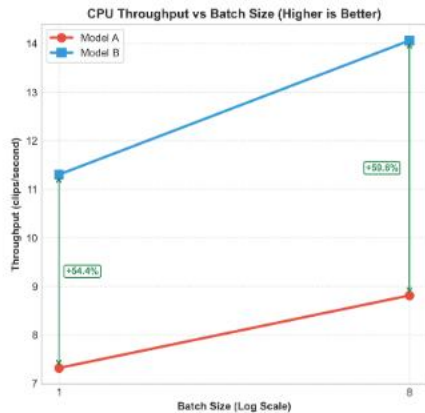
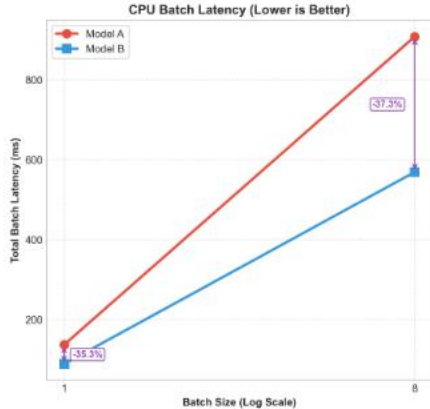
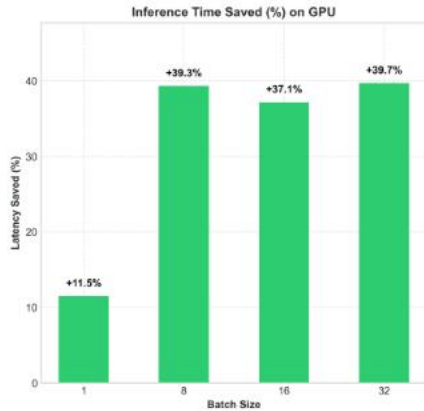
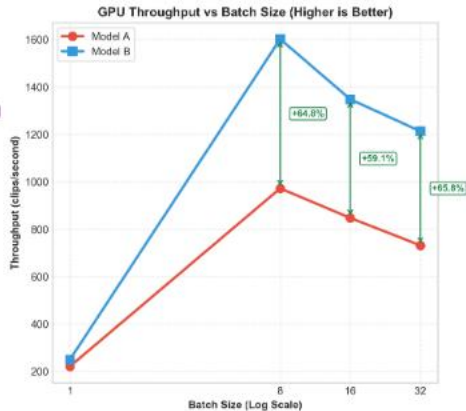
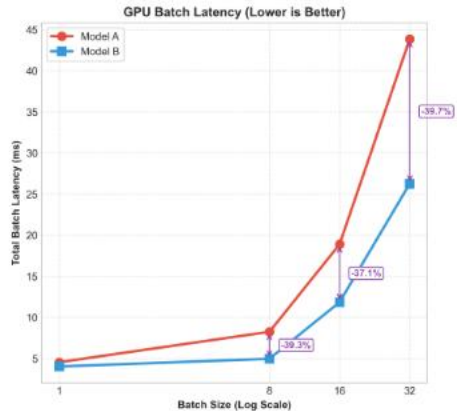
ACCURACY ACROSS GENERATIONS



Cross-Frame Distillation

Asymmetric temporal sub-sampling ($T=20$, $a=0.7$) — 16f Student vs 24f Teacher

Test set inference Benchmark (GPU & CPU): Model A (24f) (Top-1: 65.85%) vs Cross Frame Model B (16f) (Top-1: 62.68%)



PERFORMANCE GAP

-3.17%

65.85% (24f) → 62.68% (16f)

Minimal accuracy cost for temporal compression

INFERENCE SPEEDUP

2.11×

CPU latency halved (-52.6%)

GPU throughput boosted by +68.8%

Key Takeaways & Future Work

NO FRAME DIST. (24F)

ACCURACY

65.85% +6.37 pp

vs. 59.48% Baseline

CPU LATENCY

53.52 ms

24-frame computational load

GPU THROUGHPUT

991 v/s

Standard throughput

CROSS-FRAME DIST. (16F)

ACCURACY

62.68% -3.17%

vs. 65.85% Teacher

CPU LATENCY

25.36 ms -52.6% (2.11×)

Latency halved on Edge CPU

GPU THROUGHPUT

1673 v/s +68.8%

Register & Tensor Core alignment

Lessons Learned

The spatial capacity gap limits intermediate feature transfer. Logit distillation at high temperature (\$T=20\$) is the most stable guide.

Key Limitations

- Spatial capacity bottleneck of the Student.
- Over-smoothing on fast, periodic actions.
- Recursive collapse in self-distillation without oracle.

Future Work

Dynamic temperature calibration to prevent self-distillation collapse, and training on flexible temporal boundaries (variable stride).

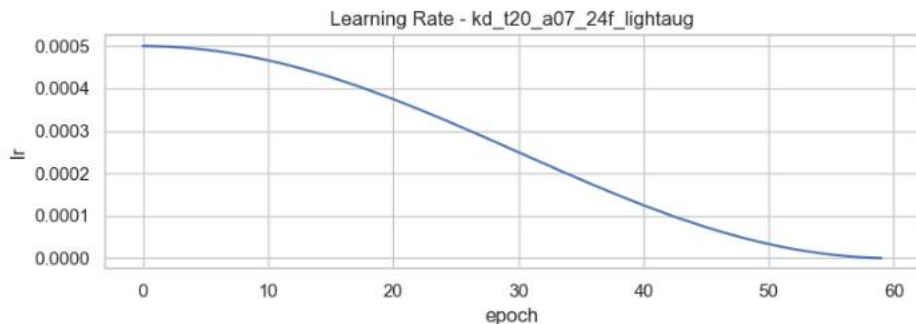
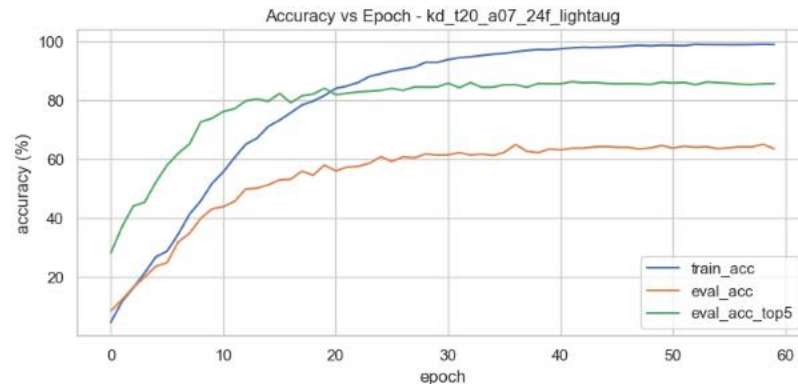
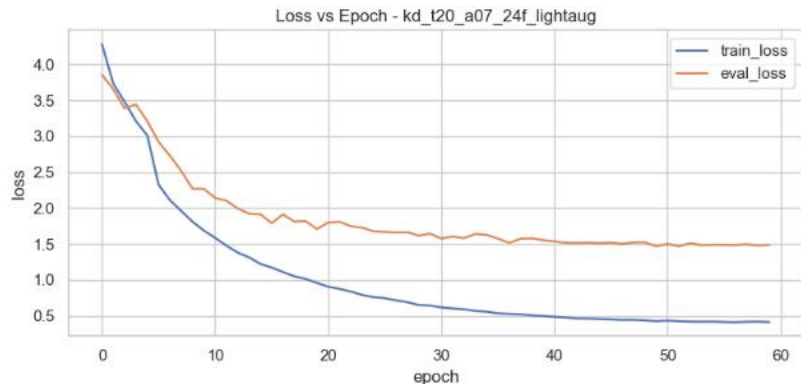
ADDITIONAL MATERIAL

Backup Slides

Quantitative plots, qualitative detailed analyses, and CM predictions.

Training Curves — KD T=20, a=0.7

Loss, accuracy, and LR schedule over 60 epochs



Stability & Convergence

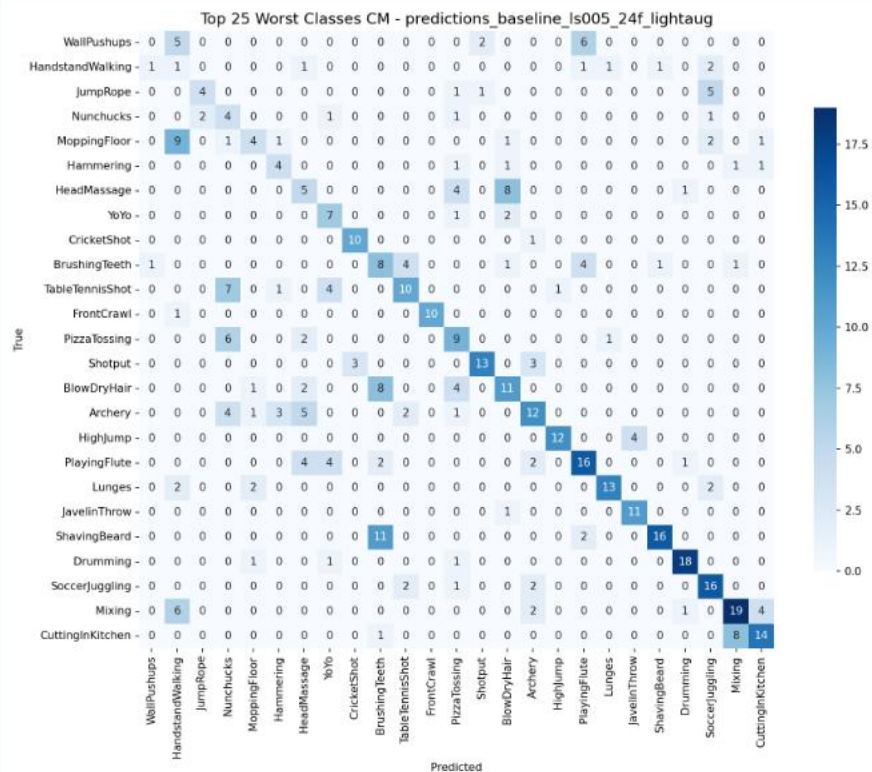
Stable training over 60 epochs. First 10 epochs serve as warmup with hard targets before soft distillation loss activates.

No strong overfitting divergence — regularization works optimally with $T=20$.

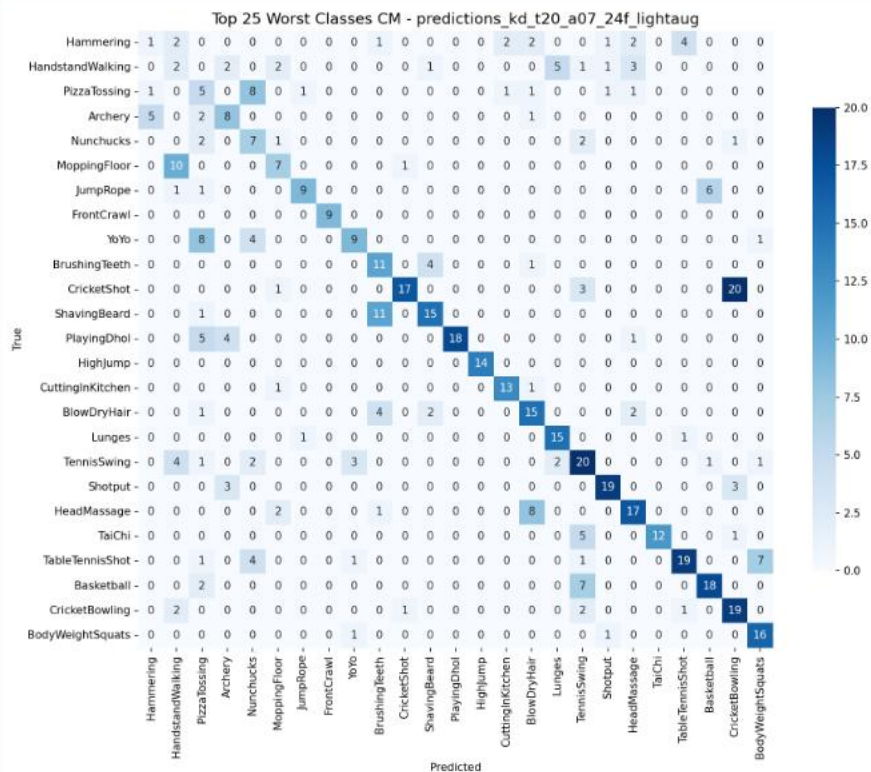
Confusion Matrix & Error Analysis

Baseline vs KD T=20 - Top 25 worst classes

BASELINE - 59.48%



KD T=20 - 65.85%



Attention Transfer Loss: Before and After

Decomposing 3D video features to resolve the spatial-temporal collapse

Before: Single Collapsed Loss

$$\mathcal{L}_{AT} = \sum_i || \text{normalize}(\text{mean}_c(F^2))_{\text{flat}} - \dots ||^2$$

- **Block Mismatch:** Hooked ResNet stages [3, 4, 5]. Stage 5 is the classifier head (produces 2D logits `[B, 101]`), which caused runtime crashes and zero gradient contribution.
- **1D Interpolation Mismatch:** Resizing a flattened 1D vector of shape `[T × H × W]` treats space and time as a single coordinate line. This mathematically mixed pixels across frame boundaries and unrelated spatial points, destroying feature geometry.
- **Temporal Signal Dilution:** Normalizing the entire flattened space meant temporal peaks were washed out by spatial dimensions.

RUN 4391

59.14% (Top-1)

Severe accuracy collapse under noisy gradients

After: Spatial & Temporal Decomposition

$$\mathcal{L}_{AT} = \beta_s \mathcal{L}_{\text{spatial}} + \beta_t \mathcal{L}_{\text{temporal}}$$

- **Correct Block Pairing:** Hooked ResNet stages [2, 3, 4] (intermediate 3D features: `[B, C, T, H, W]`), correctly matching student stages [2, 4, 6].
- **2D Bilinear Interpolation:** Spatial attention is calculated as `mean<sub>c,t</sub>(F<sup>2</sup>)` and resized spatially via 2D bilinear interpolation, preserving frame layout.
- **Temporal AT Isolation:** Temporal attention is calculated as `mean<sub>c,h,w</sub>(F<sup>2</sup>)` yielding `[B, T]`, which is transferred directly without any interpolation noise.

RUN 4601

64.58% (Symmetric) • **65.03%** (Temporal)

Successful feature alignment and positive transfer